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The association between voluntary disclosure and corporate governance in the presence of severe agency conflicts

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ABSTRACT

Agency conflicts between different types of investors are particularly severe in the presence of high family and block-holder ownership. By focusing on a setting characterised by high ownership concentration, we study the role of independent directors in promoting transparency through increased disclosure. In our tests, we use a sample of Spanish firms and, consistent with prior work, show that the presence of these directors is strongly associated with increased voluntary disclosure. Additionally, we find that when an executive director takes on Chair responsibilities the level of voluntary information is reduced, creating potential conflicts with the role of independent directors. Our results suggest that a strong legal framework holds firm-level clashes of interest in check. We conclude that this regulatory environment can create sufficient incentives to bring together the interests of minority and majority shareholders and guarantee an efficient monitoring role of independent directors. However, results suggest that other mechanisms should be reinforced in order to improve the role of governance control on agency relationships, particularly in the case of the concentration of Chair and executive responsibilities.
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1. Introduction

There is an ongoing debate on the joint role of high quality financial information and corporate governance provisions in reducing information asymmetries and ameliorating agency conflicts. Recent work puts forward arguments suggesting that these mechanisms are both substitutes (Bushman, Chen, Engel, & Smith, 2004) and complements (Ahmed & Duellman, 2007). In our paper, we contribute to this literature in two specific ways. First, we study the relationship between corporate governance and information quality from a broader perspective by focusing on voluntary disclosures. Second, we take into account that the performance of these mechanisms is greatly influenced by the legal and institutional setting in which firms operate. There is little prior evidence on how institutional factors may moderate the link between information quality and corporate governance mechanisms.

Our aim is to shed additional light on this association by focusing on a setting typified by high ownership concentration, and consequently, serious agency conflicts between controlling and minority shareholders (Shleifer & Vishny, 1997). We specifically look at the role of independent directors as a way to enhance information transparency through increased voluntary disclosure. The decision to increase this disclosure (and transparency) is predicted to act as a safeguard to the interests of minority shareholders.

Against this backdrop, we test the hypothesis that the presence of independent directors increases voluntary disclosure of information, thus protecting minority shareholders, even when there is high ownership concentration. Additionally, we look at whether the presence of a significant block-holder affects the role of independent directors. Finally, we test whether the legal framework plays a decisive role in guaranteeing the appointment of truly independent professional directors and in promoting positive complementarities between these control mechanisms.

Fama (1980) and Fama and Jensen (1983) identify outside independent directors as being essential to the effective monitoring and advising role of corporate boards. This monitoring role can be exercised in multiple ways. One method is by enhancing corporate transparency and accountability through alternative reporting devices, such as management forecasts, press releases or additional disclosures in the annual report, all of which reduce the costs inherent to the agency relationship (Healy & Palepu, 2001). The current mandatory financial disclosure model is considered imperfect as it does not always provide the information demanded by users. It is precisely these perceived short-comings in the current business model that have led professional accounting organizations and regulators to increase voluntarily disclosed information in annual reports (Beattie, McInnes, & Fearnley, 2004).

The monitoring role of independent directors may be either enhanced or compromised by certain institutional and firm-specific characteristics. The presence of a majority shareholder can prevent independent directors from performing their control role properly due to, among other reasons, the risk of collusion between the majority shareholder and the independent director (Patelli & Prencipe, 2007). As Cheng and Jaggi (2000) argue, the appointment of independent directors in family-controlled firms may be influenced by personal ties that affect

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their independence and in turn, their ability to improve disclosure and effective monitoring. However, it can also be argued that companies with either a high concentration of outside ownership or those that are family-controlled are more likely to appoint highly respected independent professionals to improve transparency and the firm's reputation in order to reduce the costs of the agency relationship that exists between majority and minority shareholders (Shleifer & Vishny, 1997). In any case, we expect the legal framework and enforcement mechanisms protecting minority shareholders to play a significant part in this relationship, guaranteeing the appointment of highly qualified independent professionals and creating mechanisms that make information more transparent in firms with high ownership concentration.

Most of the prior empirical literature in this area has looked at either Anglo-Saxon or Asian countries. Little research has been done on other continental European countries (Babio & Muiño, 2005; Patelli & Prencipe, 2007) where institutional differences, particularly in ownership structures and legal enforcement mechanisms, may lead to significant variations in the reported complementarities between corporate boards and information disclosures in the governance process.

Spain is an interesting framework in which to test these complementarities, because it is characterised by high ownership concentration and a significant proportion of listed family-controlled firms (Faccio & Lang, 2002). Family block-ownerships and "dominant" shareholders are commonly present in listed Spanish firms, where the latter control an average of 30% of the share capital. At the same time, the recent change[1] in this legal framework has not only promoted transparency in listed firms but also guaranteed the presence and independence of non-executive directors. In addition, in line with the idea that agency conflicts are particularly severe in the chosen setting, prior evidence on the effectiveness of independent directors in Spain has offered mixed results, suggesting that independent boards may have fallen short in their monitoring role (García Osma & Gill de Albornoz, 2004).

Spain is therefore in a good position to contribute to the debate on whether the independent directors' monitoring role is impaired or enhanced in the context of high ownership concentration (Patelli & Prencipe, 2007). Detailed information on a firm's ownership and governance structures can be manually collected through the Spanish Corporate Governance Code (CGC henceforward) which requires the identification of non-executive directors in two separate categories: gray[2] and independent directors. Additionally, financial disclosure requirements have been traditionally less specific than in other countries (i.e. the UK or the US[3]) allowing firms more discretion and the freedom to identify the main determinants of disclosure.

Based on a sample of 62 listed Spanish firms, we create an unweighted hand-collected voluntary disclosure index based on 76 items related to the information disclosed in the annual reports. The reduced size of the Spanish capital market allows us to create a self-constructed index, thus avoiding sample selection bias related to analysts' disclosure indexes. Following prior work, together with the proportion of independent directors we control for other governance variables: the size of the board of directors, the doubling up of executive and Chair responsibilities, the degree of ownership concentration and the existence of a significant block-ownership. We also look into other relevant firm-specific determinants of voluntary disclosure.

Empirical results confirm that even in a context of high ownership concentration, with a relatively significant presence of blockholder share capital, independent directors affect the quantity of voluntary information disclosed among listed firms. Therefore, capital concentration does not outweigh the role of independent directors, whose presence enhances transparency and accountability through reporting information beyond that required by accounting regulations. The results not only contribute to the literature and debate on the complementarities between information and governance mechanisms in the agency relationship, but also suggest the need to develop strong legal and enforcement safeguards that guarantee the appointment of genuinely independent directors. In fact, results suggest that even in a strong regulatory environment, the effectiveness of governance mechanisms should be periodically tested to assess potential improvements. The empirical analysis reveals that, in spite of the effect that independent directors have on disclosure, the duality of executive and Chair responsibilities negatively affects transparency, creating potential conflicts with independent directors.

We contribute the existing literature on this topic by looking at the complementary role of independent and voluntary corporate disclosure in a context where agency problems are severe. In particular, we focus on a setting typified by high ownership concentration, where the conflict of interests between minority and majority shareholders may limit the monitoring role of independent directors and therefore, the beneficial complementarities between governance mechanisms and financial information.

The remainder of this paper is organized as follows. Section 2 reviews the prior literature on corporate governance and voluntary disclosure and formulates the research hypotheses. Section 3 describes the data collection, sample selection procedure and introduces the information requirements for corporate boards. Finally, Sections 4 and 5 describe the research method and results. Section 6 concludes.

2. Corporate governance and voluntary disclosure: developing the hypotheses

2.1. Independent directors and disclosure

A good corporate governance system is a key element in optimising the performance of a business in the best interests of shareholders, limiting agency costs and favoring the survival of corporations (Fama & Jensen, 1983). The board of directors is one of the most important internal controls where external independent directors play a key role in shareholders' interests, "*carrying out tasks that involve serious agency problems*" between managers and shareholders (Fama, 1980; Fama & Jensen, 1983).

From this premise, since the beginning of the 90s,[4] an increasing number of countries have started to work on the development of CGCs to promote confidence in financial reporting and governance mechanisms in a context of increasing globalization of capital markets, where small investors have been gaining importance. Following academic and professional recommendations, CGCs refer to two main categories of directors: executive and independent non-executive directors. While the former have the knowledge and expertise on how the firm is run, the latter play an advising and monitoring role. Non-executive directors are determinant in reducing the costs of the agency relationship. However, due to the relevance of ownership participation on corporate

[1] The enactment of the Transparency Act in 2003 (26/2003) reinforced transparency and information requirements on corporate boards. Since its enactment, firms are required to file a corporate governance report, giving detailed information on their boards' structure. Boards must comply with the recommendations of the CGC.

[2] Rosenstein and Wyatt (1990, p. 235) define gray as outside directors "*family members of insiders, attorneys whose firms represent the firm, investment or commercial bankers whose firms have relationships with the firm, consultants to the firm and directors who personally or through their employers have substantial business dealings with the firm*". Gray directors are the non-executive directors representing majority shareholders while independent directors represent small investors' interests.

[3] As Luo, Courtneay, and Hossain (2006) explain, Verrecchia (2001) suggests that due to the rich US disclosure environment, empirical studies on disclosure based on US firms are unlikely to discover substantial first order effects of voluntary disclosure on information asymmetry.

[4] Following the publication of the UK Cadbury Report in 1992, the majority of the developed countries published similar Codes of Conduct dealing with the structure of the boards of directors. 1994: Canada; 1995: Australia, France and the European Union; 1996: The Netherlands; 1997: Japan and EE.UU.; 1998: Spain, Belgium Germany and Italy; 1999: Greece, Ireland and Portugal. 2000: Denmark. 2001: Sweden; 2002: Austria; 2003: Finland and New Zealand; 2004: Norway. The European Corporate Governance Institute offers an overview and free access to all the Corporate Governance Codes around the world. http://www.ecgi.org/codes/all_codes.php.

boards, countries with a high ownership concentration (i.e. Spain) include two separate categories of non-executive directors in their CGC: gray and independent.

In this context, the traditional agency conflict is outweighed by the conflicts that arise between large vs. minority shareholders. Concentrated ownership is considered a governance mechanism in itself, as large owners may use their power to appoint independent non-executive directors to control managerial decisions effectively (Shleifer & Vishny, 1997). However, large shareholders dominate the "*decision control*" role in the company and their interests may not always be the same as those of minority shareholders which leads to an additional agency problem within the firm. Under circumstances where majority shareholders may be incentivized to expropriate wealth from minority shareholders,[5] independent directors play a twofold control role; they can prevent this expropriation not only from large shareholders but also from managers.

The legal environment plays an important part in guaranteeing independence on corporate boards together with other "*good*" governance mechanisms (Shleifer & Vishny, 1997). It is for this reason, following theoretical academic recommendations, that most CGCs require a majority of non-executive independent directors so that they can fulfill their controlling role without interference from insiders or majority shareholders.

There is extensive empirical evidence on how independent directors use their control role in varying institutional settings.[6] A recent stream of research looks at the complementarities of both corporate governance and disclosure, focusing on the role of independent directors as a mechanism to enhance transparency and disclosure. As Lim et al. (2007) explain, one outcome of effective governance is greater accountability, and implicitly, more voluntary disclosure of information. Most of the empirical studies corroborate this assertion, finding positive complementarities between independent directors and disclosure[7] (Babio & Muiño, 2005; Bujaki & McConomy, 2002; Cheng & Courtenay, 2006; Cheng & Jaggi, 2000; Donnely & Mulcahy, 2008; Lim et al., 2007; Patelli & Prencipe, 2007).

However, these positive complementarities might not be applicable to all institutional settings.[8] As previously explained, most of the current literature focuses on either Asian or Anglo-Saxon countries but little research has been done in continental Europe where ownership structures and legal enforcement mechanisms differ considerably and may affect the governance–disclosure relationship.

2.2. Independence among directors and the role of the institutional setting

Outside directors' lack of independence severely limits their ability to perform their role efficiently. Independence can be significantly curtailed in a context of serious agency conflicts brought about by high ownership concentration. In fact, the presence of majority shareholders is claimed to be among the main constraints due to the risk of complicity between themselves and the appointed independent director (Patelli & Prencipe, 2007). Appointing new directors is a board decision, however, large shareholders have the power to influence this decision and, thus impair the board's independence. In addition, they may favor the appointment of independent directors based on their personal relationships rather than on professional expertise.

The want of real independence, together with lacking financial expertise, are claimed to be the explanation for independent directors' lack of effectiveness when majority, insider or family owners appoint directors[9] (Park & Shin, 2004). This might also explain the absence of positive complementarities between governance and disclosure.

The legal framework plays a significant role in moderating the complementarities that the governance and company's information mechanisms may achieve. Guaranteeing the appointment of highly qualified independent professionals and placing safeguards to enhance information transparency in tightly controlled firms strengthens the complementarities between voluntary disclosure and governance mechanisms. In this kind of environment, managers and majority shareholders face higher costs associated to wealth expropriation, and misleading or self-serving disclosures.

Independent directors belong to an "*intensive monitoring package*" (Ho & Wong, 2001) promoted at the institutional level that persuades companies not to withhold value relevant information. Independent and qualified professionals will always reinforce transparency and perform their monitoring role efficiently to uphold their reputation in the labor market. In a strict regulatory environment focused on strengthening transparency, firms will want to increase their reputation for transparency (Patelli & Prencipe, 2007) and appoint independent directors who will perform their advising and monitoring role efficiently.

Under the Spanish Code of Corporate Governance (CNMV, 2006), independent directors are "*appointed for their professional and personal qualities*" that allow them to "*perform their duties without being influenced by any connection with the company, its shareholders or its management*". Independent directors must be appointed by the nomination committee. The presence of this committee on corporate boards guarantees the appointment of suitable directors and in turn, the efficient performance of the board itself (CNMV, 2006). The nomination committee can only be formed by external directors of whom the majority must be independent. Even though the Spanish legal regulations leave companies free to decide whether to follow governance recommendations, all listed firms must report and explain any deviation from the recommendations.[10] In fact, the enactment of the 26/2003 law on transparency[11] requires listed firms to prepare an annual Corporate Governance Report where companies provide detailed information on their governance structure and compliance with the CGC recommendations.

In addition to regulatory characteristics, previous evidence on Spain reveals that the conflict of interests between minority and majority shareholders (Shleifer & Vishny, 1997) only occurs at a much higher level of ownership concentration when compared to Anglo-Saxon countries[12] (De Miguel et al., 2004). Our empirical analysis is therefore

[5] Shleifer and Vishny (1997) offer detailed explanation of the different ways in which wealth is expropriated from the different types of claim holders within the firm.

[6] A number of empirical papers corroborate that outside independent directors represent minority shareholder interests well (Lim, Matolcsy, & Chow, 2007). Prior literature also corroborates a positive impact of independent directors on a firm's performance (Erhardt, Verberl, & Shrader, 2003; Rosenstein and Wyatt, 1990); controlling earnings management practices (Klein, 2002; Peasnell, Pope, & Young, 2000, 2005; Xie, Davidson, Wallace, & DaDalt, 2003); limiting financial fraud (Beasley, 1996) or on certain company transactions where serious agency problems may arise (Agrawal & Knoeber, 1996; Brickley, Coles, & Terry, 1994; Brickley & James, 1987; Weisbach, 1988).

[7] Certain authors such as Eng and Mak (2003) or Gul and Leung (2004) observe a negative relationship. However, this is/may be due to/attributed to their definition of external directors because, as Cheng and Courtenay (2006) explain, they do not divide between gray and independent directors. In spite of the caveats and limitations on the measurement of voluntary disclosure (Lim et al., 2007) most of the empirical literature highlights the role of independent directors as a determinant explanatory factor for the higher levels of voluntary disclosure results.

[8] Authors as Ho and Wong (2001) do not find a significant relationship between independent non-executive directors and voluntary disclosure, questioning the independence of directors on Hong Kong corporate boards.

[9] Other authors have claimed that independent directors have limited involvement and knowledge of the firm's daily operations and can therefore be easily misled by inside executive directors (Lim et al., 2007). However, to counter this argument is the stance that because of their concern about receiving misleading information they request additional voluntary information to protect their professional reputation and avoid litigation from minority shareholders (Fama & Jensen, 1983; Lim et al., 2007).

[10] However, even though listed companies can choose whether to comply with the CGC, their reporting must refer to the concepts used in the official report — i.e. firms must adhere to the Code's definition of "independent" director when explaining the composition of the board. If the director does not meet the minimum CGC requirements, they cannot be considered independent (CNMV, 2006).

[11] The cited law additionally promotes transparency and accountability policies requiring listed firms to operate a website in order to provide investors with updated news and value relevant information. Directors are responsible for keeping this information updated.

[12] De Miguel, Pindado, and De la Torre (2004) reveal that the value of Spanish firms rises until ownership concentration reaches 87%.

performed in an institutional context where minority shareholders are legally protected and the appointment of independent directors is guaranteed. This creates a "strong governance environment" that forces majority shareholders to appoint independent directors who carry out their monitoring role efficiently, strengthening the complementarities between governance and information. Because of this, in spite of the high ownership concentration and blockholder ownership, we expect positive complementarities between governance and disclosure, which means higher levels of voluntary disclosure in companies with a higher proportion of independent directors. The current legal framework guarantees the appointment of independent directors that enhance information transparency to reduce the costs of information asymmetries in a context of high ownership concentration.

We formulate the following hypotheses:

Hypothesis 1. *Ceteris paribus: There is a positive complementary relationship between independent directors and the extent of voluntary disclosure.*

2.3. The role of additional corporate governance characteristics on corporate disclosure

As Gul and Leung (2004) argue, the role of corporate governance on the agency relationship between managers and shareholders is best examined by looking at several corporate governance mechanisms. We therefore control for three additional governance characteristics in the empirical model: the doubling up of executive and Chair responsibilities, the ownership structure and board size.

Separating the position of CEO and chairman of the board avoids a conflict of interests and helps to improve the monitoring function of the board (Jensen, 1983). Therefore, when the chairman and the CEO functions fall onto the same person – CEO duality – the concentration of too much power in one person may compromise the monitoring role of the board (Forker, 1992) and affect the quality and amount of information disclosed. Authors like Donnely and Mulcahy (2008), Ho and Wong (2001) and Gul and Leung (2004) document a negative relationship between CEO duality and voluntary disclosure.

A detailed analysis of Spanish governance characteristics reveals that in listed firms there is a significant proportion of CEOs with Chair responsibilities and to a lesser extent, executive directors doubling up as chairman.[13] As we will observe later, this situation affects 71% of the sample firms which means that only 29% appoints either a gray or independent director as chairman. Given this context, we consider both the concentration of the CEO/Chair or executive/Chair responsibilities as Duality. We assume that in both cases there is an overconcentration of responsibilities and therefore, a potential risk of compromising the oversight role of the board. Based on the theory postulates and previous empirical evidence we expect to find a negative relationship between the degree of voluntary disclosure and the concentration of the CEO/executive and Chair responsibilities in the same person (Duality).

The firm's ownership structure is associated with different levels of disclosure (Gelb, 2000). More specifically, information disclosure is expected to increase where ownership is more spread out (Raffournier, 1995) and where minority shareholders require greater transparency and information. However, majority shareholders may also want disclosure increased due to capital market pressures or for other reasons (Salter, 1998). When strong regulatory mechanisms are in place, firms are interested in achieving a reputation for being highly transparent (Patelli & Prencipe, 2007) to avoid losing investors. In this light, the presence of these shareholders can imply greater disclosure and this can also be said for large blockholders, those major shareholders controlling a significant portion of the company shares. However, most of the empirical evidence reports a negative[14] relationship between ownership concentration and voluntary disclosure (Babio & Muiño, 2005; Chau & Gray, 2002; Cheng & Courtenay, 2006; Cheng & Jaggi, 2000; Patelli & Prencipe, 2007). Based on previous results in the Spanish context, we expect a negative relationship between voluntary disclosure and this governance control variable.

The Unified Spanish Governance Code (CNMV, 2006, pp 14) states that board size "*has a bearing on its efficiency and on the quality of decision-making*". While we expect larger boards to increase board monitoring capabilities, this benefit may be reduced by poorer communication and decision-making associated to larger groups (John & Senbet, 1994). As Jensen (1983) argues, small boards are more effective in monitoring the CEO, limiting the possibility of taking opportunistic decisions. In fact, previous empirical results reveal a negative relationship between board size and firm value (Yermack, 1996). Cheng and Courtenay (2006) document the absence of a significant relationship between board size and voluntary disclosure for a sample of Singapore firms. However, the relationship between firm and board size (Denis & Sarin, 1999) together with the tendency for big firms to be under greater pressure from stakeholders to provide information suggest that larger boards are inclined to disclose more.

3. Sample selection and data collection

3.1. Sample selection and measurement of voluntary disclosure

The final sample consists of 62 non-financial Spanish companies listed on the Madrid Stock Exchange in 2005. The quantity of voluntary disclosure is measured using an unweighted[15] disclosure index, computed using a binary coding scheme that identifies the presence or absence of the different information items considered. The voluntary disclosure index has been computed based on hand-collected data from the fiscal year 2005 annual reports. Our empirical analysis is based on the first year of the IAS/IFRS adoption, where both the importance of additional disclosures and good governance practices were reinforced with the aim of achieving more efficient capital markets with high quality accounting standards.

Even though companies have alternative ways to report additional voluntary information,[16] studies like Botosan (1997) or Lang and Lundholm (1993) see a direct link between annual report disclosure and alternative ways of presenting corporate information. The annual report is one of the main sources of corporate information and the main source of data[17] in the voluntary disclosure empirical literature.

With an initial sample of 124 non-financial companies listed on the Madrid Stock Exchange (IBEX-35 and IGBM), we exclude companies with non-consolidated financial statements, unavailable annual reports and firms with missing information on the corporate governance structure. Finally, we exclude those companies with missing data for the control variables. Our final sample consists of 62 listed companies. Table 1 shows the final sample selection procedure (Panel A), as well as the composition of the final sample (Panel B).

[13] The current Spanish CGC does not make any recommendation on the advisability of either separating or concentrating the two positions. The Code refers to the lack of empirical evidence and the international practice divergence as the main arguments to avoid proposing a recommendation on this point (CNMV, 2006, p. 18). However, under a context of Chairman/CEO duality the Code proposes the appointment of a "*lead independent director to request the calling of board meetings or the inclusion of new business on the agenda; to coordinate and give voice to the concerns of external directors; and to lead the board's evaluation of the Chairman*" (CNMV, 2006, p. 19, 48). In the empirical analysis we do not control for the appointment of a lead independent director in companies with duality.

[14] Other authors like Donnely and Mulcahy (2008), Haniffa and Cooke (2002) or Eng and Mak (2003) do not find evidence of a significant relationship between ownership and voluntary disclosure.

[15] We do not weight the related importance of the selected items to avoid subjectivity in the index computation.

[16] Corporate websites, press releases, intellectual capital reports, corporate social responsibility reports, meetings with the financial analysts, and management forecast announcements.

[17] Previously, literature in the US context used the AIMR (*Association for Investment and Management Research*) disclosure rankings to measure the disclosure level. However, it has been claimed that these rankings are biased towards larger firms.

Table 1
Sample selection procedure and list of firms comprising the sample.

Panel A: Sample selection procedure	No.
Non-financial firms listed in the Madrid Stock Exchange in 2005	124
Not required to report consolidated financial statements	12
Reporting period different from 31st December 2005	5
Missing observations for corporate governance variables	9
Missing observations for control variables	36
Final sample	62

Panel B: List of firms comprising the sample

Company name	Company name
Logista S.A.	Amper S.A.
NH Hoteles S.A.	Befesa Medio Ambiente S.A.
Prosegur S.A.	Indra Sistemas S.A.
Service Point Solutions S.A.	Adolfo Domínguez S.A.
Sol Meliá S.A.	Altadis S.A.
Acciona S.A.	Baron de Ley S.A.
ACS Actividades Construcción y Servicios	Campofrío Alimentación
Fomento Construcciones y Contratas S.A.	Dogi International Fabrics S.A.
Obrascon Huarte Lain S.A.	Ebro Puleva S.A.
Aguas de Barcelona S.A.	Gamesa
Enagas S.A.	Grupo Empresarial Ence
Gas Natural SDG S.A.	Iberpapel Gestión S.A.
Hullas del Coto Cortes	Indo Internacional S.A.
Iberdrola S.A.	Miquel y Costas S.A.
Petroleos (Cepsa)	Papeles y Cartones de S.A.
Red Eléctrica de España	Pescanova S.A.
Repsol YPF S.A.	SOS Cuétara S.A.
Unión Fenosa S.A.	Tableros de Fibras S.A.
Fadesa Inmobiliaria S.A.	Tavex Algodonera S.A.
Inbesos S.A.	Tele Pizza S.A.
Inmobiliaria Colonial S.A.	Viscofan S.A.
Metrovacesa S.A.	Ercros S.A.
Cementos Portland Valderrivas S.A.	Construcciones y Auxiliar de Ferrocarriles, S.A.
Uralita S.A.	Duro Felguera S.A.
Antena 3 S.A.	Elecnor S.A.
Sogecable S.A.	Tubacex S.A.
Telecinco S.A.	Abertis S.A.
Acerinox S.A.	Cintra
Lingotes Especiales S.A.	CLH
Tubos Reunidos S.A.	Iberia S.A.
Abengoa S.A.	Telefónica S.A.

Our measure of voluntary disclosure is a self-constructed index based on a checklist of 76 identified information items related to seven areas of information[18]: The checklist has been created based on the following: the framework of the Steering Committee Report of the Business Reporting Research Project of the Financial Accounting Standards Board in 2001, the recommendations of the Enhanced Business Reporting Consortium (EBR) report published in 2005 and the disclosure checklists included in previous studies such as Botosan (1997), Cheng and Courtenay (2006) and Lim et al. (2007).

A dichotomous variable (1/0) has been used to identify each information item that can appear in the firm's annual report and is then used as a base from which the disclosure index is computed. The dummy variable for each item on the checklist takes the value of 1 if the company discloses information related to that item in the annual report; and 0 otherwise. Similar to previous studies, to avoid subjectivity in index computation, all the checklist items are considered to have the same relevance for the external users of information. The voluntary disclosure index (D_INDEX) is computed as the sum of all the dichotomous variable values for each company, divided by the total number of items included in the information checklist (76).

Table 2 Panel A shows descriptive statistics for the total voluntary disclosure index and sub-indexes for the 62 companies in the final sample. The mean voluntary disclosure index is 0.25, revealing that sample companies disclose about 25% of the 76 items comprising the general index. This value is higher than those reported in similar studies for other countries. Lim et al. (2007) for Australian companies, Cheng and Courtenay (2006) for Hong Kong firms and Patelli and Prencipe (2007) for Italian firms, report an average index of 0.18, 0.29 and 0.14, respectively. The D_INDEX score ranges from 0.07 to 0.48, suggesting a large variation in voluntary disclosure practices across Spanish firms.

Table 2 reveals that firms are more likely to disclose information on corporate social responsibility (I_CSR = 0.33), non-financial information (I_NFI = 0.34) and Historical information (I_H = 0.27). The I_IAS index is significantly high due to the analysis of the 2005 annual reports, where most firms in the sample reported information on the impact of adopting IAS/IFRS standards.

One of the main caveats of designing a voluntary disclosure index is that it implies a certain degree of subjectivity in administering the disclosure checklist (Cheng & Courtenay, 2006). Following Botosan (1997) and Cheng and Courtenay (2006) we assess the validity of the index for capturing disclosure levels. One of the basic validity analyses of its internal consistency is a correlation analysis of each one of the index components. As Cheng and Courtenay (2006) explain, "*disclosure strategies for a firm are expected to be similar along all avenues*", which is to say that a firm with high levels of voluntary information as reported in the general voluntary disclosure index (D_INDEX) is expected to have a high disclosure level in most of the information areas. Non-reported results of the Pearson and Spearman correlation analyses of all the sub-indexes of information reveal, not only a significant correlation with each other, but also with D_INDEX. These results corroborate the consistency of the dependent variable.

3.2. Corporate governance and control variables

As previously explained, since the enactment of the 26/2003 law on transparency, listed Spanish firms are required to prepare an annual corporate governance report, in keeping with the statutory information requirements of the national Securities and Exchange Commission (CNMV). All corporate governance variables have been collected from the 2005 corporate governance filed reports. In particular, we collect detailed information on board composition and ownership structure. Regarding board composition we measure board size and the proportion of independent, gray, and executive directors comprising the board. The analysis of the ownership structure focuses on the degree of ownership

Table 2
Descriptive statistics on the voluntary disclosure index and sub-indexes.

Variables	n	Mean	Median	Std.dev.	Max	Min
D_INDEX	62	0.2501	0.253	0.0866	0.480	0.067
I_H	62	0.2726	0.200	0.1757	0.800	0
I_CSR	62	0.3306	0.375	0.3320	1.000	0
I_IC	62	0.2247	0.214	0.1754	0.571	0
I_PRI	62	0.0355	0.000	0.0624	0.267	0
I_BCK	62	0.0327	0.294	0.1501	0.706	0
I_NF	62	0.3364	0.286	0.1937	0.857	0
I_MA	62	0.1419	0.100	0.1751	0.600	0
I_IAS	62	0.8011	1.000	0.3694	1.000	0

D_INDEX = general voluntary disclosure index; I_H = historical information disclosure index; I_CSR = corporate social responsibility disclosure index; I_IC = intangible and intellectual capital disclosure index; I_PRI = projected information disclosure index; I_BCK = background and general information disclosure index; I_NF = non-financial statistics disclosure index; I_MA = management analysis disclosure index; I_IAS = IAS/IFRS adoption disclosure index.

[18] Historical information, corporate social responsibility, intangible and intellectual capital, projected information, general information about the firm, non-financial statistics, management analysis and IAS/IFRS adoption. Appendix 1 reports the number of items included in each of the seven information areas, as well as a detailed list of the 76 items.

Table 3
Descriptive statistics on corporate governance variables.

Variables	n	Mean	Median	Std.dev.	Max	Min	No. (%)
Board size	62	11.95	11	4.01	20	5	
No. of executive directors	62	2.24	2	1.13	5	0	
No. of gray directors	62	5.40	5	3.96	19	0	
No. of independent directors	62	3.97	4	2.47	13	0	
No. of other directors	61	0.36	0	0.84	4	0	
% executive directors	62	0.20	0.19	0.12	0.63	0	
% gray directors	62	0.42	0.44	0.24	1	0	
% independent directors	62	0.35	0.33	0.19	0.82	0	
% other external directors	62	0.03	0	0.077	0,4	0	
Ownership concentration	62	51.68	56.23	22.67	97.29	0	
Main shareholder ownership	62	30.16	24.50	21.21	91.16	0	
No. of majority shareholders	62	3.69	3.5	2.084987	9	0	
Majority independent directors							16 (25.80)
Majority gray directors							25 (40.3%)
Majority executive directors							2 (0.09)
Majority external directors							60 (96.7%)
Chairman/CEO (yes/no)							44 (70.97%)
Big four	62						56 (92.32%)

concentration and the presence of a significant blockowner. The former is measured as the percentage of share capital owned by shareholders who possess more than 3% of the share capital while the latter is identified when one or a maximum of two shareholders own more than 30% of share capital.

Table 3 reports detailed information on the characteristics of corporate boards among sample firms. The board of directors (Panel B) has a mean size of 12 members, ranging from a minimum of 5 to a maximum of 20 members. The board is composed of a majority of gray directors (42%), followed by independent (35%) and executives (20%). In addition, 71% of the companies from our sample double up on the Chair and CEO/executive responsibilities. The average capital owned by majority shareholders amounts to 51.68%, with 97% as a maximum percentage of concentration. As stated in Section 1, ownership of Spanish companies is highly concentrated among a small number of shareholders. The mean number of majority shareholders is four and the principal shareholder controls an average of 30% of company shares.

The characteristics of the ownership structure are consistent with the higher presence of gray directors on the board of directors. 40.3% of the companies (25 companies) have a majority of gray directors while 25.8% of the total companies (16 companies) have a board with a majority of independent directors. Executive directors are in the minority and in only two firms are they the majority.

Financial control variables used in the empirical analysis were collected from WorldScope. Following previous literature on the determinants of voluntary disclosure (Ahmed & Courtis, 1999) we collect data related to the firm's size, leverage and profitability. Table 4 reports the descriptive statistics for the control variables which shape the main characteristics of the companies in the sample. The descriptive evidence for these variables is consistent with prior work and reveals a wide cross-sectional variation across the sample firms.

4. Method

We study the association between corporate governance characteristics and disclosure quality. For the empirical analysis we estimate the following model:

$$D_INDEX_{it} = \alpha + \sum_{j=1}^{4} \beta_j BOARD_{jit} + \sum_{q=1}^{5} \gamma_q VAR_CONTROLS_{jit} \quad (1)$$

where D_INDEX_{it} is the value of the voluntary disclosure index for each company in 2005, BOARD corresponds to the vector of corporate governance variables including: the proportion of independent directors on the board (%_IND), board size (BOARD), the doubling up of Chair and CEO responsibilities (DUALITY), and ownership concentration (CCAP), measured with a dummy variable (1–0) that takes the value of one when the main shareholders own more than 51.68%[19] of the firm. An extension of the basic vector of corporate governance variables controls for the impact of blockholder ownerships on disclosure. The variable BLOCK takes the value of 1 if one or a maximum of two significant shareholders control over 30% of the company shares. Otherwise, the BLOCK variable takes the value of 0. This variable controls for the impact of blockholder ownerships on voluntary disclosure and allows us to test the role of independent directors in enhancing information transparency even in the presence of a significant blockowner. As previously argued, the ownership structure is an important determinant of voluntary disclosure. Ownership concentration implies a lower proportion of free floating capital and therefore, reduces the need and shareholder pressures to enhance voluntary disclosure. This argument also holds good for the blockholder ownership variable.

VAR_CONTROLS corresponds to the vector of control variables. Previous empirical literature documents that corporate attributes such as size, leverage, profitability and growth opportunities are some of the main cross-sectional determinants[20] of voluntary disclosure (Ahmed & Courtis, 1999). To avoid multicollinearity problems among the control variables, due to the significant relationship between board size and company size[21] (Table 7), we do not control for company size using the total assets variable. Instead, it is indirectly controlled in the regression model both through the use of the board size variable in the BOARD vector and the market-to-book ratio in the CONTROL vector.

[19] This value corresponds to the mean value of the "capital concentration" variable for the sample firms. See Table 3.

[20] Empirical literature has looked at additional determinants of disclosure such as: (a) the audit firm size (Wallace, Naser, & Mora, 1994); (b) the internationalization of the firm, not only in commercial terms but also by their presence in international capital markets (Khanna, Palepu, & Srinivasan, 2004), (c) the use of stock option plans as a manager remuneration mechanism (Aboody & Kasnzink, 2000), and (d) media visibility (Cormier & Magnan, 2003). These additional explanatory factors have not been included in the vector of control variables as they are not significantly different across the sample firms.

[21] Size has been identified in numerous studies as the main determinant for voluntary disclosure. Authors like Meek, Roberts, and Gray (1995) or Hossain, Perera, and Rahman (1995) find that big companies are more likely to disclose information not only due to lower information production costs but also because of lower potential competitive disadvantages (proprietary costs). Disclosing more information can also be the result of pressure from external users. Agency costs are higher for companies with more outside capital (Jensen & Meckling, 1976) and the proportion of that capital tends to be higher for big firms (Leftwich, Watts, & Zimmerman, 1981). Therefore, larger firms are expected to disclose more information. However, this may significantly increase potential political, legal or competition costs (Watts & Zimmerman, 1986), particularly for big firms which tend to have greater visibility in the market, increasing the collateral effects of greater transparency.

Table 4
Descriptive statistics on control variables.

Variables	n	Mean	Median	Std.dev.	Max	Min
Total assets	62	6,327,031	1,304,084	10,694,174	64,789,100	60,170
Market capitalization	62	4,239,967	1,547,609	8,948,129	60,810,783	37,573
Shareholders' equity	62	1,436,375	393,310	2,874,395	15,262,000	29,560
Leverage (LEV)	62	1.403	0.884	1.551	7.585	0.0012
Market-to-book (MB)	62	3.340	2.30	2.815	14.876	0.835
Return on Assets (ROA)	62	0.056	0.047	0.052	0.33	−0.001
Number of analysts	54	9.79	8.77	6.92	31.25	1

Leverage (LEV) is measured as the total debt to equity ratio. ROA is measured as the Return on Assets ratio and MB is measured as the market capitalization to book value of equity ratio.

Highly leveraged firms bear more agency costs due to the potential wealth transfers from debtholders to shareholders (Jensen & Meckling, 1976), creating a need to disclose more information to improve transparency and communication with their creditors (Meek et al., 1995). However, the empirical evidence relating to the impact of leverage on voluntary disclosure is inconclusive (Ahmed & Courtis, 1999) as are the results of the impact of profitability on voluntary disclosure (Ahmed & Courtis, 1999). Authors like Meek et al. (1995) argue that highly profitable companies disclose more information to show their superior performance. However, other authors find either a non-significant or negative relationship between disclosure and performance (Cheng & Courtenay, 2006; Gul & Leung, 2004; Hossain et al., 1995; Raffournier, 1995). A more detailed view of the relationship between performance and disclosure is offered by Lang and Lundholm (1993) who suggest that the relationship between profitability and greater disclosure is only positive for companies with greater information asymmetries between managers and investors. A similar argument applies for the market-to-book ratio (MB), representing the firm's growth expectations. Authors such as Gul and Leung (2004) or Lim et al. (2007) argue that companies with higher growth potential need to disclose more information in order to signal to the market that the stock value is not "overvalued" and reduce uncertainty about future financial performance.

The economic sector is an additional documented explanatory factor of voluntary disclosure. Companies acting in the same sector are inclined to adopt similar informative practices and topics which affect disclosure. Because our sample is broadly spread over 21 economic sectors,[22] with no concentration of companies in any particular sector, we have not included any control variable in the econometric analysis to identify potential industry differences in disclosure.

To avoid endogeneity problems that could affect the results in the empirical analysis, we use a two stage least square procedure econometric approach similar to previous empirical literature (Cheng & Courtenay, 2006; Gul & Leung, 2004; Lim et al., 2007). We estimate first the values of the main corporate governance variable (%_IND) and use the estimated value (%_IND_est) in the second stage regression as an independent variable of the model.

The following model has been used for the estimation of the proportion of independent directors (%_IND):

$$\%_IND_{it} = \alpha + \beta_1 BOARD_{it} + \beta_2 CAP_{it} + \beta_3 LASSET_{it} + \beta_4 LEV_{it} + \beta_5 ROA_{it} + \beta_6 MB_{it} + \varepsilon_{it} \quad (2)$$

BOARD is the size of the board of directors; CAP represents the total stake of the firm's capital owned by majority shareholders[23]; LASSET represents the size of each company measured as the logarithm of total assets; LEV is the leverage ratio measured as total debt over total equity; ROA is the economic profitability of the company and finally, the market-to-book ratio (MB) measures the potential for company growth.

Based on the theoretical postulates and empirical results in previous studies (see Linck, Netter, and Yang (2008) for references to the main literature on the determinants of the board structure) we expect a positive and significant relationship between the dependent variable (%_IND) and all the explanatory variables except for CAP and BOARD. Based on previous evidence, higher ownership concentration implies the presence of a higher proportion of gray and executive directors on the board, representing the interests of the dominant shareholders. Additionally, the definition of the dependent variable as the proportion of independent directors on the total board implies a negative relationship with the size of the board (BOARD). The expected relationship between LASSET, LEV, ROA, MB and the dependent variable (%_IND) is positive. Big companies tend to have higher ownership dispersion (Leftwich et al., 1981) and greater cash flows, making it necessary to recruit a larger number of independent non-executive directors to effectively oversee managers (Boone, Casares Field, Karpoff, & Raheja, 2007). Similarly, highly profitable companies or those with high growth expectations are not only more attractive to independent directors (Lim et al. (2007) but they also suffer from higher information asymmetries that require the presence of independent directors to promote transparency between dominant and minority shareholders. Finally, highly leveraged firms are expected to have a significant presence of independent directors to promote transparency and strengthen creditors' confidence.

Table 5 shows the summary of statistics for the first stage regression results. All the estimated coefficients are statistically significant except the profitability variable (ROA). Size (LASSET), leverage (LEV) and the

Table 5
Summary statistics from the ordinary least squares regression. Stage 1 regression — relationship between the proportion of independent directors and firm specific characteristics.

$\%_IND_{it} = \alpha + \beta_1 BOARD_{it} + \beta_2 CAP_{it} + \beta_3 LASSET_{it} + \beta_4 LEV_{it} + \beta_5 ROA_{it} + \beta_6 MB_{it} + \varepsilon_{it}$

Dependent variable = %_IND

Variables	Expected sign	Coef.	T-stat	Pr > |t|
Intercept		0.04983	0.25	0.8057*
BOARD	−	−0.02619	−3.83***	0.0003##
CAP	−	−0.46397	−5.4***	<.0001###
LASSET	+	0.05819	3.33***	0.0016
LEV	+	−0.02429	−1.3#	0.1993
ROA	+	−0.17295	−0.36	0.72
MB	+	0.01939	2.1**	0.0405
Adj R-Sq	0.4254			
F-stat (p value)	<.0001			

%_IND = proportion of independent directors in the board of directors. BOARD = board size. CAP = ownership concentration measured as the proportion of the firm's capital owned by significant shareholders. We consider as significant a capital share over 3%. LASSET = logarithm of total assets. LEV = total debt to equity ratio. ROA = return on assets. MB = market-to-book ratio.
10% significance — one-tailed T-test.
5% significance — one-tailed T-test.
1% significance — one-tailed T-test.
* 10% significance — two-tailed T-test.
** 5% significance — two-tailed T-test.
*** 1% significance — two-tailed T-test.

[22] We use the CNMV (Comisión Nacional del Mercado de Valores) industry sector classification.
[23] We consider as majority shareholder a capital share over 3%.

MB ratio are positively related to the proportion of independent directors on the board. Conversely, higher capital concentration (CAP) and the size of board (BOARD) have a negative impact on the dependent variable. The adjusted R^2 coefficient reaches a value of 0.425 which is slightly higher than those reported in other studies like Lim et al. (2007).

5. Analysis and discussion of results

5.1. Descriptive and univariate analysis

Table 6 reports a descriptive analysis of the disclosure differences according to corporate governance and firm-specific characteristics. The 62 sample companies have been divided in two groups based on the average value of each discriminant variable. To assess the role of independent directors (%_IND), sample firms have again been divided in two groups. The first comprises companies with a proportion of independent directors below the average reported value in Table 3 while the second is composed of those above the average reported value. Results show a significantly higher value of the voluntary disclosure index for the second group of companies and partially confirm the role of independent directors as an important control mechanism to improve information transparency.

While independent directors improve the level of voluntary disclosure, the presence of a higher proportion of executive directors on the board seems to have the opposite effect. In these cases, companies have a lower value of the voluntary disclosure index, although the difference is not statistically significant. These results suggest a negative relationship consistent with the argument that the presence of executive directors reduces information transparency. Results on the role of gray directors are not significant either, although firms with a greater proportion of these directors have a higher value in the voluntary disclosure index.

Table 6
T-test of differences in means on D_INDEX, based on corporate governance and firm-specific characteristics. Wilcoxon non-parametric statistic has been used to test for the differences in the discrete DUALITY and Block variables.

Variables		n	D_INDEX	T-stat	PR > |t|
%_independent directors	<mean	38	0.2375	−1.45#	*0.1522*
	>mean	24	0.27		
%_gray directors	<mean	30	0.2458	−0.38	*0.7064*
	>mean	32	0.2542		
%_executive directors	<mean	36	0.2593	0.98	*0.3316*
	>mean	26	0.2374		
Total assets	<mean	48	0.2333	−3.00***	*0.0039*
	>mean	14	0.3076		
Ownership concentration	<mean	26	0.2723	1.74*	*0.0863*
	>mean	36	0.2341		
Board size	<mean	32	0.2308	−1.85*	*0.0699*
	>mean	30	0.2707		
LEV	<mean	42	0.2441	−0.79	*0.4352*
	>mean	20	0.2627		
ROA	<mean	40	0.243	−0.87	*0.388*
	>mean	22	0.263		
DUALITY	1	44	0.2396	1.77*	*0.0764*
	0	18	0.2755		
Block	0	51	0.2507	−0.03	*0.9705*
	1	11	0.246		

LEV = total debt to equity ratio. ROA = return on assets. BLOCK = dummy variable that takes the value of one if one (maximum two) significant shareholders control over 30% of the company shares. Otherwise, the Block variable takes the value of 0. DUALITY = dummy variable that takes the value of 1 when the chairman and CEO responsibilities lie on the same person. Otherwise, this variable takes the value of 0.

10% significance — one-tailed T-test.
5% significance — one-tailed T-test.
1% significance — one-tailed T-test.
* 10% significance — two-tailed T-test.
** 5% significance — two-tailed T-test.
*** 1% significance — two-tailed T-test.

When using total assets as a discriminating variable, results are consistent with the size of the firm as one of the main determinants of voluntary disclosures. Bigger companies disclose considerably more information. Similar results are reported for board size. Furthermore, results for capital concentration as a discriminating variable are consistent with the idea that ownership concentration implies higher managerial and majority shareholder control, reducing information disclosure and transparency. Results reveal a statistically significant lower value of the D_INDEX variable for companies with high capital concentration. However, there is no longer a statistical significance if we look at the differences in disclosure based on the presence of a blockholder ownership (BLOCK) as discriminating variable. Finally, the doubling up of CEO and Chair responsibilities does significantly affect the level of disclosure.

The correlation analysis of the D_INDEX and the control variables reported in Table 7 corroborate the results from the descriptive analysis. The correlation matrix shows Pearson (upper half) and Spearman (bottom half) correlation coefficients for all test variables.

Correlation coefficients of the D_INDEX variable with the control variables show a statistically significant correlation with BOARD, CAP, DUALITY, %_IND and %_EJE. These results are consistent with the descriptive analysis in Table 6, supporting the relationship between voluntary disclosure, board composition and ownership structure. The Pearson and Spearman correlation coefficients of D_INDEX with board size (BOARD) are significantly positive, indicative of a higher degree of voluntary disclosure for companies with a greater number of directors on their boards. Capital concentration (CAP) has significantly negative Spearman and Pearson correlation coefficients suggesting less disclosure as ownership concentration increases. Furthermore, the DUALITY variable has a significantly negative (Spearman) correlation with D_INDEX.

The correlation of the percentage of independent directors (%_IND) with D_INDEX is only statistically significant for the Pearson correlation coefficient, although in both cases it is positive, indicative of higher levels of voluntary disclosure in the presence of independent directors.

The variable %_IND is negatively correlated with CAP. These variables have a negative and statistically significant correlation coefficient which points to a negative relationship between ownership concentration and the proportion of independent directors. %_IND is also highly negatively correlated with BOARD and %_DOM. These results ratify the presence of endogeneity across the board composition variables, confirming the need for a two-stage least square regression econometric procedure.

The control variables (LEV, MB and ROA) also significantly correlate with each other, suggesting potential multi-collinearity problems. However, as reported in Table 8, the Variance Inflation Factor is not higher than 2.5 for any of the variables in the model.

5.2. Regression results

The second stage regression uses the estimated dependent variable (%_IND_est) as one of the explanatory variables of the model. As a sensitivity analysis, the consistency of the results is tested using an alternative dependent variable, RD_INDEX. This variable represents the transformation of the D_INDEX variable in deciles, measuring relative levels of disclosure, using a procedure similar to those in Cheng and Courtenay (2006) and Botosan (1997).

Table 8 summarizes the multiple regression results. Panel A reports the results of D_INDEX as a dependent variable. Panel B reports the results of RD_INDEX as the dependent variable. Four different regression models have been run based on the following equation:

$$\frac{\text{D_INDEX}_{it}}{\text{RD_INDEX}_{it}} = \alpha + \beta_1 \text{BOARD}_{it} + \beta_2 \%\text{_IND_est} + \beta_3 \text{DUALITY}_{it} + \beta_4 \text{CCAP}_{it} + \beta_4 \text{Block}_{it} + \beta_5 \text{LEV}_{it} + \beta_6 \text{ROA}_{it} + \beta_7 \text{MB}_{it} + \varepsilon_{it} \quad (3)$$

Table 7
Pearson and Spearman correlation matrix of D_INDEX, corporate governance and control variables.

		Pearson correlation coefficients											
		D_INDEX	MB	SIZE	LEV	ROA	BOARD	%_IND	%_DOM	%_EJE	CAP	BLOCK	DUALITY
Spearman correlation coefficients	D_INDEX	1	−0.0165	0.3627**	0.1123	0.0192	0.2526**	0.2117*	−0.0574	−0.2157*	−0.2419*	−0.0153	−0.1895
			0.8986	*0.0038*	*0.3848*	*0.8821*	*0.0476*	*0.0986*	*0.6576*	*0.0923*	*0.0582*	*0.9059*	*0.1402*
	MB	0.1454	1	0.1333	0.4209***	0.3097**	0.1980	0.0229	0.0218	0.0009	0.2234	0.1480	−0.2542**
		0.2596		*0.3017*	*0.0007*	*0.0143*	*0.1229*	*0.8600*	*0.8665*	*0.9944*	*0.0809*	*0.2510*	*0.0462*
	SIZE	0.3068**	0.3107**	1	0.3585**	−0.0597	0.6698***	0.0861	0.0640	−0.1985	0.0318	0.0055	−0.0150
		0.0153	*0.0140*		*0.0042*	*0.6450*	*<.0001*	*0.5058*	*0.6212*	*0.1219*	*0.8063*	*0.9661*	*0.9079*
	LEV	0.1963	0.3451***	0.4832***	1	−0.3901***	0.0666	0.0026	−0.0732	0.1779	0.1495	0.2618**	−0.2323*
		0.1263	*0.0060*	*<.0001*		*0.0017*	*0.6071*	*0.9838*	*0.5717*	*0.1664*	*0.2462*	*0.0398*	*0.0693*
	ROA	0.1244	0.1535	0.0257	−0.5193***	1	0.1400	−0.0381	0.1315	−0.1354	0.0904	0.0430	−0.1334
		0.3353	*0.2336*	*0.8428*	*<.0001*		*0.2779*	*0.7689*	*0.3084*	*0.2939*	*0.4846*	*0.7399*	*0.3012*
	BOARD	0.2423*	0.3045**	0.6668***	0.1748	0.2994**	1	−0.2344*	0.3793***	−0.3675***	0.1110	−0.1748	−0.0435
		0.0578	*0.0161*	*<.0001*	*0.1742*	*0.0181*		*0.0667*	*0.0024*	*0.0033*	*0.3904*	*0.1742*	*0.7370*
	%_IND	0.1973	0.0470	0.0434	−0.0799	−0.0017	−0.2598**	1	−0.8404***	0.0071	−0.5590***	0.0056	0.0716
		0.1243	*0.7167*	*0.7378*	*0.5371*	*0.9894*	*0.0415*		*<.0001*	*0.9561*	*<.0001*	*0.9657*	*0.5804*
	%_DOM	−0.0146	−0.0112	0.1097	0.0064	0.1542	0.3935***	−0.8284***	1	−0.4366***	0.4183***	−0.1853	−0.1330
		0.9103	*0.9314*	*0.3962*	*0.9606*	*0.2315*	*0.0016*	*<.0001*		*0.0004*	*0.0007*	*0.1493*	*0.3028*
	%_EJE	−0.2077	−0.0313	−0.2225*	−0.0416	−0.2471*	−0.3808***	0.0694	−0.4306***	1	0.1021	0.1614	0.1983
		0.1053	*0.8092*	*0.0822*	*0.7482*	*0.0529*	*0.0023*	*0.5920*	*0.0005*		*0.4299*	*0.2100*	*0.1223*
	CAP	−0.2272*	0.1966	0.0056	0.0137	−0.1007	0.0378	−0.4473***	0.3332***	0.0585	1	0.3294***	−0.1186
		0.0758	*0.1257*	*0.9654*	*0.9157*	*0.4360*	*0.7708*	*0.0003*	*0.0081*	*0.6517*		*0.0089*	*0.3584*
	BLOCK	−0.0059	0.1309	0.0318	0.0766	−0.1828	−0.1528	0.0626	−0.1841	0.1830	0.3362***	1	−0.2092
		0.9636	*0.3104*	*0.8059*	*0.5536*	*0.1549*	*0.2355*	*0.6286*	*0.1519*	*0.1544*	*0.0075*		*0.1028*
	DUALITY	−0.2278*	−0.1628	−0.0516	−0.0794	−0.1172	−0.0419	0.0607	−0.1610	0.2535**	−0.0834	−0.2403*	1
		0.0749	*0.2061*	*0.6903*	*0.5395*	*0.3645*	*0.7465*	*0.6394*	*0.2114*	*0.0468*	*0.5193*	*0.0600*	

D_INDEX = general voluntary disclosure index. MB = market-to-book ratio. SIZE = logarithm of total assets. LEV = total debt to equity ratio. ROA = return on assets. BOARD = board of directors' size. %_IND = proportion of independent directors in the board. %_DOM = proportion of gray directors in the board. %_EJE = proportion of executive directors in the board. CAP = ownership concentration measured as the proportion of the firm's capital owned by the main shareholders. BLOCK = dummy variable that takes the value of one if one (maximum two) significant shareholders control over 30% of the company shares. Otherwise, the Block variable takes the value of 0. DUALITY = dummy variable that takes the value of 1 when the Chair and CEO responsibilities lie with the same person. Otherwise, this variable takes the value of 0. Numbers in italics correspond to the 2-tailed significance probability.

* 10% significance — two-tailed T-test.

** 5% significance — two-tailed T-test.

*** 1% significance — two-tailed T-test.

Table 8
Summary statistics from the two stage least squares regression. Stage 2 regression — relationship between the voluntary disclosure variable and the vectors of board and control variables, using the fitted value of %_IND (%_IND_est).

$D_INDEX_{it}/RD_INDEX_{it} = \alpha + \beta_1\ BOARD_{it} + \beta_2\ \%_IND_est + \beta_3 DUALITY_{it} + \beta_4\ CCAP_{it} + \beta_4\ LEV_{it} + \beta_5\ ROA_{it} + \beta_6\ MB_{it} + \varepsilon_{it}$

Panel A: Dependent variable = D_INDEX

		Model 1		Model 2		Model 3		Model 4	
		D_INDEX		D_INDEX		D_INDEX		D_INDEX	
		Coefficient	T-stat	Coefficient	T-stat	Coefficient	T-stat	Coefficient	T-stat
Intercept		0.1178	1.60	0.1138	1.5	0.0940	1.25	0.0922	1.19
BOARD	+	0.0076***	2.69	0.0078***	2.67	0.0086***	2.86	0.0087***	2.79
%_IND_est	+	0.2100*	1.95	0.2100*	1.97	0.2441**	2.15	0.2448**	2.14
DUALITY	−	−0.0391*	−1.74	−0.0368#	−1.52	−0.0405*	−1.7	−0.0396#	−1.58
CCAP	−	−0.0067	−0.24	−0.0084	−0.3	−0.0044	−0.02	−0.0011	−0.04
Block	+			0.0084	0.27			0.0041	0.13
LEV						0.0114	1.2	0.0111	1.14
ROA	+					0.2058	0.78	0.2014	0.75
MB	+					−0.0088*	−1.76	−0.0888	−1.74
Adj R-Sq		15.01%		17.15%		15.45%		16.99%	
Max. VIF		1.99		1.99		2.23		2.23	
F-stat		3.69		2.92		2.59		2.23	
(p value)		(*0.0096*)		(*0.0206*)		(*0.0222*)		(*0.0395*)	

Panel B: Dependent variable = RD_INDEX

		Model 1		Model 2		Model 3		Model 4	
		RD_INDEX		RD_INDEX		RD_INDEX		RD_INDEX	
		Coefficient	T-stat	Coefficient	T-stat	Coefficient	T-stat	Coefficient	T-stat
Intercept		0.0395	1.00	0.0401	0.99	0.0256	0.64	0.0270	0.65
BOARD	+	0.0034	2.25**	0.0034	2.18**	0.0040	2.49**	0.0039	2.37**
%_IND_est	+	0.0806	1.41#	0.0806	1.39#	0.1031	1.70*	0.1026	1.68*
DUALITY	−	−0.0244	−2.03**	−0.0247	−1.91*	−0.0259	−2.04**	−0.0266	−1.99**
CCAP	−	−0.0042	−0.29	−0.0040	−0.26	0.0004	0.03	0.0009	0.05
Block	+			−0.0013	−0.08			−0.0030	−0.18
LEV	+					0.0051	1.01	0.0053	1.01
ROA	+					0.1183	0.84	0.1216	0.85
MB	−/+					−0.0048	−1.81*	−0.0049	−1.80*
Adj R-Sq		11.21		13.69		11.69		10.08	
Max VIF		1.99		1.99		2.23		2.23	
F-stat		2.92		2.30		2.15		1.85	
(p value)		(*0.286*)		(*0.0569*)		(*0.0532*)		(*0.0872*)	

BOARD = board size. %_IND_est = proportion of independent directors on the board as estimated in the 1st stage regression. DUALITY = dummy variable (1–0) that takes the value of one when the Chair and CEO responsibilities double up. CCAP = ownership concentration measured with a dummy variable (1–0) that takes the value of one when the main shareholders own more than 40% of the firm. BLOCK = dummy variable that takes the value of one if one (maximum two) significant shareholders control over 30% of the company shares. Otherwise, the Block variable takes the value of 0. LEV = total debt to equity ratio. ROA = return on assets. MB = market-to-book ratio. D_INDEX = general voluntary disclosure index. RD_INDEX corresponds to the transformation of the D_INDEX variable in deciles. RD_INDEX takes values from 1 to 10.

10% significance — one-tailed T-test.
5% significance — one-tailed T-test.
1% significance — one-tailed T-test.
* 10% significance — two-tailed T-test.
** 5% significance — two-tailed T-test.
*** 1% significance — two-tailed T-test.

Results reported in Table 8 support Hypothesis 1. That is, in spite of focusing on an institutional context with serious agency conflicts, a higher proportion of independent directors increases transparency through the disclosure of information beyond the mandatory requirements. Regression coefficients for the explanatory variable %_IND_est are positive and statistically significant in all the regression models. These results are consistent when controlled not only for the impact of ownership concentration, but also for duality and in the presence of blockholder capital ownerships.

The coefficient of determination (adjusted-R^2) ranges between 15 and 17%, indicating that a moderate percentage of the variation in Y can be explained by variations in the set of independent variables. In addition, the results on the F-statistic allow us to reject the hypothesis that all the explanatory variable coefficients are simultaneously equal to zero.

Four out of the eight explanatory variables are statistically significant across the eight regression models: Board size (BOARD), Duality, market-to-book ratio (MB) and the estimated variable %_IND_est, representing the estimated value of the percentage of independent directors on the board. The signs of the regression are consistent with expectations in all cases. The results on the main governance variable %_IND_est support the main hypothesis. As reported in Table 8 (Panel A), the regression coefficients are significant at the 5% level in models 3 and 4 (10% in models 1 and 2).

Results suggest that in spite of the potential agency conflicts that may arise in an institutional setting with high ownership concentration, the legal environment guarantees the appointment of genuinely independent directors that represent the information interests of minority and majority shareholders alike. In fact, results on the ownership concentration and blockownership variables do not affect the level of disclosure. The CCAP and Block regression coefficients are significant in none of the models where these variables are included.

Conversely, the Duality variable is negative and statistically significant in all cases, indicating that the concentration of the Chair and executive responsibilities significantly reduces the voluntary information disclosed by companies. Results for the BOARD variable are consistent with previous empirical studies revealing that firms with larger boards disclose more voluntary information.

Coefficients on the control variables LEV and ROA are not significant. The only exception is the MB variable where the results suggest that companies with higher growth potential avoid voluntary disclosure in order to keep strategic data from competitors.

These results document positive complementarities between governance and information mechanisms even in the presence of high ownership concentration and blockholder ownerships. These findings imply that developing a legal mechanism that promotes transparency and strong governance mechanisms strengthens the complementarities between both controlling mechanisms, offsetting the potential incentive for majority shareholders to withhold relevant information to minority shareholders. However, results also suggest that other governance mechanisms should be reinforced to improve control over the agency relationship. In particular, doubling up the responsibilities of chairman and executive director is seen to significantly reduce the level of voluntary information, creating potential conflicts with the role of independent directors.

6. Summary and conclusions

This article looks at the role of independent directors in a context of high ownership concentration, where family and blockholder ownership is very much present in listed firms. In particular, it aims to understand the complementarities between the information and governance mechanisms in an institutional setting where a conflict of interests between majority and minority shareholders may arise. Under these circumstances, majority shareholders could undermine the controlling role of independent directors and it is in this case, that the legal framework holds the key to guaranteeing the appointment of genuinely independent directors who can perform their role efficiently.

We focus on the voluntary dimension of disclosure as a key complement to mandatory financial information as a means to improve accountability and transparency in firms. A self-constructed non-weighted disclosure index is used to measure the degree of this voluntary disclosure and we assess the role of independent directors in promoting disclosure in a final sample of the 62 listed firms. Spain is an interesting arena to test for the governance-information complementarities as the current regulatory framework enhances transparency and limits the undermining of director independence. Additionally, it is characterised by high ownership concentration and the presence of family and blockholder ownerships.

The econometric analysis isolates the effect of ownership concentration on voluntary disclosure. We find that the role of independent directors is not impaired and that their presence is significantly correlated with higher levels of voluntary disclosure. Our results highlight the relevance of guaranteeing independence among board members in order to steer the interests of minority and majority shareholders towards more accountability and transparency so as to reduce the information asymmetries that arise in an open corporation.

However, regression results for the DUALITY variable suggest the need to revise the current recommendations on certain governance measures such as the concentration of Chair and CEO responsibilities that not only affects transparency but may also interfere with the role of independent directors. As previously stated, the current Spanish CGC makes no recommendation on the advisability of either separating or combining the two positions. The Report's committee justifies their decision on the lack of empirical evidence on the benefits and drawbacks of the Chair/CEO duality. However, our empirical analysis in the Spanish context corroborates a negative impact on disclosure.

The research findings highlight the relevance of the legal framework in creating a strong governance environment that represents both the disclosure interests of minority and majority shareholders and guarantees an efficient monitoring role of the board. However, evidence reveals that this framework, and in particular, the effectiveness of the governance recommendations, should be watched carefully.

There are several limitations to this study. First, we have not focused on disclosure quality but quantity. Although some previous studies have concentrated on both (Cheng & Courtenay, 2006), we have not assessed quality in order to minimise subjectivity in the index computation. In addition, sample size is limited, since the empirical results are based on a single country and in one single year. However, due to the laborious nature of the data collection to build the disclosure index, most previous literature is also characterised by a small sample sizes.

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Appendix 1. Information items

Panel A: Information categories

Category	N° items
Historical information	10
Corporate social responsibility	3
Intangibles and intellectual capital	14
Projected information	15
Background information	17
Non-financial information	7
Management analysis	5
NIC/NIIF adoption	3
Total	76

Panel B: checklist of the 76 information items related to the seven areas of information

Category
Historical information
ROE — figure or growth percentage (YES/NO)
ROE — figure or growth percentage (additional information)
ROA — figure or growth percentage (YES/NO)
ROA — figure or growth percentage (additional information)
EPS — figure or growth percentage (YES/NO)
EPS — figure or growth percentage (additional information)
Sales — figure or growth percentage (YES/NO)
Sales — figure or growth percentage (additional information)
Price per share (PPS) — figure or growth percentage (YES/NO)
Price per share (PPS) — figure or growth percentage (additional information)
Corporate social responsibility
GRI indicators (YES/NO)
Description of social programs and strategy (YES/NO)
Quantitative information on social investment (YES/NO)
Intangibles/intellectual capital
Intellectual capital report (YES/NO)
Human capital: training programs (YES/NO)
Human capital: training programs (total investment)
Human capital: training programs (number of programs)
Human capital: training programs (number or percentage of employees attending the training programs)
Human capital: employee turnover (YES/NO)
Relational capital: customer loyalty index (YES/NO)
Relational capital: customer satisfaction index (YES/NO)
Structural capital: quality certifications (YES/NO)
Structural capital: quality certifications (number)
Structural capital: investment on research (YES/NO)
Structural capital: investment on research (figure)
Structural capital: investment on development (YES/NO)
Structural capital: investment on development (figure)

Appendix 1 (*continued*)

Panel B: checklist of the 76 information items related to the seven areas of information
Category
Projected information
Descriptive information on projected sales (YES/NO)
Quantitative information on projected sales (YES/NO)
Quantitative information on projected sales (additional information)
Descriptive information on projected earnings (YES/NO)
Quantitative information on projected earnings (YES/NO)
Quantitative information on projected earnings (additional information)
Descriptive information on projected R&D expenditures (YES/NO)
Quantitative information on projected R&D expenditures (YES/NO)
Quantitative information on projected R&D expenditures (additional information)
Descriptive information on projected market share (YES/NO)
Quantitative information on projected market share (YES/NO)
Quantitative information on projected market share (additional information)
Descriptive information on projected cash flows (YES/NO)
Quantitative information on projected cash flows (YES/NO)
Quantitative information on projected cash flows (additional information)
Background information
Objectives — descriptive information (YES/NO)
Objectives — quantitative information (YES/NO)
Macroeconomic environment — descriptive information (YES/NO)
Macroeconomic environment — quantitative information (YES/NO)
Legal and political environment — descriptive information (YES/NO)
Legal and political environment — quantitative information (YES/NO)
Competitive environment — descriptive information (YES/NO)
Competitive environment — quantitative information (YES/NO)
Financial markets — descriptive information on the capital markets' general trend (YES/NO)
Financial markets — quantitative information on the capital markets' general trend (YES/NO)
Descriptive information on the company stock evolution on financial markets (YES/NO)
Quantitative information on the company stock evolution on financial markets (YES/NO)
Detailed information on ownership structure (YES/NO)
Information about the management stock ownership (YES/NO)
Detailed information on management remuneration (YES/NO)
Information on good corporate governance practices (YES/NO)
Information about meetings with financial analysts (YES/NO)
Non-financial information
Number of employees
Information on the company contracting policy (YES/NO)
Information on the distribution of employees by gender (YES/NO)
Information on the distribution of employees by age (YES/NO)
Information on average compensation per employee (YES/NO)
Information on number of units sold (figure or growth percentage) (YES/NO)
Information on market share (YES/NO)
Management analysis
Management analysis of changes in net sales (YES/NO)
Management analysis of changes in the level of expenditures (YES/NO)
Management analysis of changes in earnings (YES/NO)
Management analysis of changes in market share (YES/NO)
Management analysis of changes in R&D expenses (YES/NO)
NIC/NIIF adoption
Descriptive information on the main effects of the adoption of NIC/NIIF (YES/NO)
Quantitative information-reconciliation — main effects of NIC/NIIF adoption on shareholders' equity (YES/NO)
Quantitative information-reconciliation — main effects of NIC/NIIF adoption on earnings (YES/NO)

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